MIL-T-82329C
25 August 1969
SUPERSEDING
MIL-T-82329B (TD)
3 May 1968

MILITARY SPECIFICATION
TRAINERS, ELECTRONICS APPLICATION

This specification is mandatory for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers electronics application trainers, (see 6.1).

1.2 Classification.- The trainers shall be of the following types, (see 6.1.1):

(a) Type I - Student Participation Trainer

(b) Type II - Instructor Demonstrator Trainer

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

W-C-596	Connector, Plug, Electrical; Connector Receptacle, Electrical
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Fiberboard
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof, For Packaging

Military

MIL-R-11	Resistor, Fixed, Composition, (Insulated), General Specification For

FSC 6940

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MIL-C-25	Capacitor, Fixed, Paper-Dielectric, Direct Current (Hermetically Sealed in Metallic Cases), General Specification For
MIL-P-116	Preservation, Methods of
MIL-S-3950	Switches, Toggle, General Specification For
MIL-L-10547	Liners, Case and Sheet, Overwraps, Water-Vaporproof, Flexible
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	Electronic Equipment, Naval Ship and Shore, General Specification
MIL-C-39003	Capacitor, Fixed, Electrolytic, Tantalum, Solid-Electrolyte, Established Reliability, General Specification For

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-749	Preparation and Submission of Data For Approval of Nonstandard Electronic Parts

DRAWINGS

Naval Training Device Center(NAVTRADEVCEN)

DL 1002	Reference Data for Device 6B17 and 6B17A

PUBLICATION

Naval Training Device Center (NAVTRADEVCEN)

335-101-6B17	Test Procedures and Readings

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Procuring Contracting Officer.)

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3. REQUIREMENTS

3.1 Materials.- Materials not referred to in the reference data of 3.9 shall be selected on the basis of compatibility with the established design and suitability to the specific application in meeting the performance requirements specified in 3.4.

3.1.1 Control knobs for Type I trainers shall be of black bakelite, 1 1/4 inch size, pointer-type with metal inserts and hexagon-socket headless setscrews. Control knobs for Type II trainers shall be of black bakelite, 2 1/4 inch size, pointer-type with metal inserts and hexagon-socket headless screws.

3.2 Design.- The trainer shall be in accordance with the reference data of 3.9 and as specified in 3.2.1 through 3.10.

3.2.1 Interchangeability.- All parts, subassemblies and assemblies having the same part number shall be directly and completely interchangeable with each other in respect to installation and performance, regardless of sources from which obtained, without alteration and without selection for fit or performance.

3.2.2 Safety.-

3.2.2.1 Grounding.- The trainers shall be equipped with a 6-foot, 3-wire (one of which is a grounding wire) line cord and a grounding-type plug conforming to W-C-596 suitable for the power required. The ground wire shall not be interrupted by any power-switching devices inserted into the line cord.

3.2.3 Component layout.- Component layout shall be in accordance with the reference data of 3.9.

3.3 Construction.-

3.3.1 Panels.-

(a) Type I panels shall be of standard aluminum 13 by 17 by 4 inches with removable bottom plates

(b) Type II panels shall be standard aluminum 36 by 42 by 6 inches with removable bottom plates

(c) Type I and II trainers shall be identical except for physical size. Electrical components shall be identical and located at the appropriate point in the front panel diagram.

3.3.2 Power supply mounting.- The power supply shall be mounted on an aluminum chassis and constructed using reference data of 3.9 as a guide.

3.3.3 Soldering.- Solder and soldering techniques shall be in accordance with requirement 5 of MIL-STD-454.

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3.4 Performance.-

3.4.1 Controls and components.- All controls and movable components shall operate smoothly without binding. The controls and components, when set to a selected position, shall not reset to a different position until moved manually.

3.5 Details of components.- The trainers shall consist of the following major components, the details of which shall be as shown in reference data of 3.9 and as specified in 3.5.1 through 3.5.7:

(a) Panel 1 - Basic Resistance

(b) Panel 2 - Complex Resistance

(c) Panel 3 - Capacitance - Inductance

(d) Panel 4 - RC Time Constant

(e) Panel 5 - Series LCR

(f) Panel 6 - Series Resonance

(g) Panel 7 - Parallel LCR

(h) Panel 8 - Parallel Resonance

3.5.1 Power supply.- The power supply for panels 1, 2, 3 and 4 shall be in accordance with the reference data of 3.9. The power supply for panel 1 shall provide outputs of 20 volts and 40 volts DC at a load of 10 milliamperes. The power supply in panel 2 shall provide 40 volts DC at a load of 10 milliamperes. The power supply for panels 3 and 4 shall provide 20 volts DC at a load of 10 milliamperes.

3.5.1.1 The voltages (20v/40v) from the power supply for panel 1 shall be selectable by means of a switch mounted on the front panel. These voltages shall be obtained by appropriate taps on the secondary of the power transformer.

3.5.2 Wire.- Wire shall be number 20 AWG in accordance with MIL-E-16400. Color code red for voltage leads and black for ground leads shall be used.

3.5.3 Capacitors.- Capacitors other than electrolytic shall be within 10 percent tolerance and shall comply with MIL-C-25. Capacitors C401, C402 and C403 in Panel 4 shall be solid dielectric in accordance with MIL-C-39003.

3.5.4 Electrolytic capacitors.- Electrolytic capacitors shall be high quality commercial grade of selected tolerance to meet the performance requirements within the tolerance specified in 3.6.

3.5.5 Fixed resistors.- Fixed resistors shall be in accordance with

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MIL-R-11 within ± 5 percent tolerance.

3.5.6 Toggle and rotary switches.- Toggle switches shall comply with MIL-S-3950. Rotary switches shall be Centralab PA 1000 series or equal.

3.5.7 Approval of nonstandard electronic parts.- Application for Government approval of nonstandard electronic parts for use in training equipment shall be in accordance with MIL-STD-749.

3.6 Dimensions and tolerances.- Dimensions and tolerances shall be as specified in 3.3.1, 3.5 and the reference data of 3.9. Tolerances not specified or shown shall be such, as to assure meeting the interchangeability requirements of 3.2.1 and performance requirements of 3.4. Electrical tolerances shall be plus or minus 10 percent.

3.7 Finish and color.- Finish and color of the chassis shall be etched or anodized natural aluminum. The front of each panel shall be painted gray and shall comply with MIL-E-15090, Type III, Class 2.

3.7.1 The chassis and covers shall be free of burrs or sharp edges.

3.8 Nameplates or product markings.-

3.8.1 Nameplate.- A nameplate conforming to the design and identification requirements specified by the procuring activity (see 6.2) shall be affixed to the top center of each panel of both Type I and Type II trainers.

3.8.2 Marking of components.- Identification of terminal connections, test points, and reference symbol designations shall conform to the reference data of 3.9 and shall be located adjacent to each part and inside the chassis and readily visible for maintenance purposes.

3.8.2.1 Front panel diagrams and lettering for Type I trainers shall be as shown in the reference data of 3.9 and shall be etched, photoengraved or silk-screened on the chassis front. Line width of diagrams shall be one-sixteenth of an inch. Reference symbol numbers shall be one-quarter inch in height. Front panel diagrams and lettering for Type II trainers shall be etched, photoengraved or silk-screened on the chassis front. Line width of diagrams and reference symbol numbers shall be reproduced 2 1/2 times the size for Type I trainers using the reference data of 3.9 as a guide.

3.8.2.2 Color code of resistors on panels 1 and 2 shall be marked on the outline drawing on the front panel in accordance with Radio Manufacturers Association (RMA) color code.

3.9 Government-furnished reference data.- The Government will furnish the following NAVTRADEVCEN reference data:

DL1002	Reference data for Device 6B17 and 6B17A

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3.10 Workmanship.- The trainers shall be constructed and finished with particular attention given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, etching, anodizing, silk-screening, sanding, grinding, painting, riveting, machine-screw assemblage, welding and brazing.

3.10.1 Cleanup.- Units shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign material after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection

4.3 In-Process inspection.- In-process inspection shall include such visual, electrical, and mechanical examinations and testing of materials, subassemblies, parts and accessories (including purchased standard items) during the manufacturing process as may be necessary to assure conformance to the requirements of this specification.

4.4 Quality conformance inspection.-

4.4.1 Preproduction inspection.- Preproduction inspection shall be conducted on the preproduction model and shall consist of all the examinations and tests specified in 4.5 and 4.6.

4.4.2 Production inspection.- Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government-approved preproduction model.

4.5 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

(a) Materials

(b) Construction

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(c) Safety

(d) Dimensions and tolerances

(e) Finish and color

(f) Nameplates or product markings

(g) Workmanship

(h) Preparation for delivery.

4.6 Tests.- Electrical operating tests shall be performed and results recorded in accordance with test requirements specified in NAVTRADEVCEN 335-101-6B17.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be Level A or C (see 6.2).

5.1.1 Level A.- For Level A, each trainer shall be preserved and packaged in accordance with MIL-P-116, Method II.

5.1.2 Level C.- Trainers shall be preserved and packaged to a degree which will afford protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity. This level in general, will conform to applicable carrier rules and regulations. Suppliers commercial practices may be used when they meet the requirements of this level.

5.2 Packing.- Packing shall be Level A, B or C (see 6.2).

5.2.1 Level A.- Units preserved and packaged in accordance with 5.1.1 shall be packed in exterior type shipping containers conforming to PPP-B-601. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance with the Appendix thereto.

5.2.2 Level B.- Units preserved and packaged in accordance with 5.1.1 shall be packed in exterior type shipping containers conforming to PPP-B-636. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Containers shall be sealed with tape conforming to PPP-T-60 in accordance with the Appendix thereto.

5.2.3 Level C.- Packages which require over-packing for acceptance by the carrier, shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet consolidated freight classification

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rules and regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Type I trainers will be used in a classroom for basic electronics training, in conjunction with an instruction manual. Type II trainers will be used by instructors for demonstration purposes.

6.1.1 The trainers have been designated by the Naval Training Device Center as: Device 6B17 for Type I Trainer and 6B17A for Type II Trainer, for Navy procurement.

6.2 Ordering data.- Porcurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Type trainer being procured (see 1.2)

(c) Procuring activity designation (device number or other)

(d) Nameplate design identification (see 3.8.1)

(e) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2)

6.3 Definitions.- Except as indicated in 6.3.1 and 6.3.2, definitions will be furnished upon written request to the procuring activity.

6.3.1 Preproduction model.- A model suitable for complete evaluation of mechanical and electrical form, design and performance. It shall be of final mechanical and electrical form, employ standard or nonstandard parts (approved by the agency concerned) and be completely representative of the final equipment.

6.3.2 Production unit.- A unit in its final mechanical and electrical form of final production design made by production tools, jigs, fixtures and methods.

Custodian:
Navy-TD
Army-EL
Air Force-70

PREPARING ACTIVITY:
Navy-TD
Project No. 6940-0025

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM, APR 62, 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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